TWELVE LONELY RUNNERS

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Abstract. The Lonely Runners Conjecture asserts that for any number of runners moving at distinct constant speeds around a unit circle from a common starting point, each runner is lonely at some time. A runner is said to be lonely if its distance from every other runner exceeds $1/n$, where n is the total number of runners. Tao showed that it suffices to verify the conjecture for finitely many positive integers [8]. Using computational methods, Trakultongchai verified the conjecture for up to ten runners [10]. In this work, I prove that the twelve Lonely Runners Conjecture holds if the eleven Lonely Runners Conjecture holds.

Introduction

The Lonely Runners Conjecture, attributed to Wills [11], states that for any $n > 2$, if n runners are running with constant speed around a unit circle, each runner would be eventually *lonely*. Formally that means for each runner i, there is a point in time t such that no runner is in distance $\frac{1}{n}$ of runner i. By using the symmetry of the problem, one may fix a runner to be stationary and measure all other speeds relative to it. This yields an equivalent formulation as presented in Conjecture 1.

Conjecture 1. *Let $v_1, \ldots, v_n$ be positive real numbers. Then there exists a real number t such that*

$$\|tv_i\| \geq \frac{1}{n+1} \quad \textit{for all } i = 1, \ldots, n,$$

The conjecture is trivial for 2 and 3 runners. The case of $n \leq 7$ were proven using elementary methods [4]. However, as the number of runners increases, the complexity grows exponentially. While showing the proof for the case of six runners, Bohman et al. showed that it is enough to considered integer speeds [2], reducing the problem from a continuous to a discrete one. The seven runners problem was solved in 2008 by Barajas and Serra [1] using prime filtering and number theory. Moreover, Tao showed that it is possible to computationally verify the conjecture by only considering integer speeds up to $n^{O(n^2)}$ [8], turning an infinite problem into a verifiable one. Later on, Malikiosis et al. reduces the search space drastically by showing that it is enough to consider integer speeds up to only $\binom{n+1}{2}^{n-1}$ [3].

Building on the result, Rosenfeld proved the Lonely Runners Conjecture for 8 runners [6] and Trakultongchai improved the computational search to assert for 9 and 10 runners [10].

I improved the computational method used by Rosenfeld and Trakultongchai to conclude that if the conjecture holds for 11 runners, it also holds for 12 runners.

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Rosenfeld's and Trakultongchai's Computational Method

By the result of Corollary 3 of [6], if the $k-1$ Lonely Runners Conjecture holds, the k lonely runners can be proven by verifying that there are no integral speed set $V := \{0, v_1, \ldots, v_k\}$ that is a counterexample to the conjecture for

$$v_1 \ldots v_k < B_k \quad \text{where} \quad B_k := \left(\frac{\binom{k+1}{2}^{k-1}}{k} \right)^k$$

Rosenfeld introduced the idea of breaking down the bound by generating multiple small proofs, each of which is a proof that if there is a counterexample, then $p \mid v_1 \cdots v_k$. By accumulating enough proofs for a set of primes $P := \{p_j\}$, it follows that

$$\prod_{p \in P} p \,\Big|\, v_1 \cdots v_k.$$

which shows $v_1 \cdots v_k > \prod_{p \in P} p > B_k$

In the work of Trakultongchai, he defined proper and improper solution sets and showed that if the set of improper speed sets $I(k, l, p)$ is empty for some value of l, then $p \div v_1 \cdots v_k$ in any counterexample. He argued that such l would need to be a multiple of $k+1$. [10]

Trakultongchai then proposed an efficient method that allows *lifting* a candidate solution set modulo lp into a candidate solution set modulo clp. Allowing computation to use prime factors of $k+1$ to help speed up the process. For example, with $k = 8$, the idea allows splitting the computation into 3 steps, computing $I(k, 1, p)$ then $I(k, 3, p)$ and then $I(k, 9, p)$.

Main Work

The framework introduced in [10] has a bottleneck in the first phase of computation, the calculation of $I(k, 1, p)$. Notice that for a prime modulus p, the objects of interest are speed sets whose elements lie in the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$.

Using Trakultongchai's terminology, the sets denoted $I(k, 1, p)$ represent the configurations of speeds that could potentially violate the Lonely Runners Conjecture modulo p. The following definition 2 is a reformulation of definition 3.1 from [10].

Definition 2. *Let p be a prime number greater than $k+1$ and $(p, l) = 1$. Let $\pi : \mathbb{Z}/pl\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ be the natural projection. A speed set $v := \{v_1, \ldots, v_k\}$ is such that $v_i \in \pi^{-1}((\mathbb{Z}/lp\mathbb{Z})^\times)$ for all i. A speed set v is improper modulo lp, write $v \in I(k, l, p)$ if both of the following two conditions holds:*

(1) for any $t \in \mathbb{Z}/lp\mathbb{Z}$ there exists i such that $\left\| \frac{tv_i}{lp} \right\| < \frac{1}{k+1}$.
(2) for all i, $\gcd(lp, v_1, \ldots, v_{i-1}, v_{i+1}, \ldots, v_k) = 1$.

Intuitively, the condition states that for every time parameter t modulo p, there is a runner who fails to achieve the required separation distance $\frac{1}{k+1}$ from the stationary runner. Thus the set is a potential obstruction to the Lonely Runners Conjecture when considered modulo p.

Lemma 3. *For any improper speed set $\{v_1, \ldots, v_k\}$ in $I(k, 1, p)$, there exists an improper set $\{1, u_2, \ldots, u_k\}$ in $I(k, 1, p)$ and a number $u \in \mathbb{Z}/p\mathbb{Z}$ such that $\{u, uu_2, \ldots, uu_k\} = \{v_1, \ldots, v_k\}$.*

Proof. Let $u = v_1$ and $u_i = v_1^{-1} v_i$ for each i. Then, $uu_i = v_1 \cdot v_1^{-1} v_i = v_i$. Hence

$$\{u, uu_2, \ldots, uu_k\} = \{v_1, v_2, \ldots, v_k\}.$$

It remains to verify that $\{1, u_2, \ldots, u_k\}$ is also an improper set. Since $l = 1$, $\gcd(p, v_i) = 1$ for any i, so the gcd condition holds trivially. Therefore, it enough to show the first condition. To do that, pick any $t \in \mathbb{Z}/p\mathbb{Z}$ and set $t' = tv_1^{-1}$. Since $\{v_1, \ldots, v_k\}$ is an improper set, there exists some index i such that

$$\frac{1}{k+1} > \left\| \frac{t'v_i}{p} \right\| = \left\| \frac{tv_1^{-1}v_i}{p} \right\|$$

If $i = 1$ then $\|t/p\| < \frac{1}{k+1}$ otherwise $\|tu_i/p\| < \frac{1}{k+1}$ In either case the defining condition holds for the set $\{1, u_2, \ldots, u_k\}$, proving that it is also an improper set. □

The lemma shows an equivalent relation of improper sets in $I(k, 1, p)$ under multiplication by units. Consequently, when enumerating possible counterexamples modulo p it suffices to consider only normalized sets of the form $\{1, u_2, \ldots, u_k\}$. This reduces the search space by a factor of p.

When lifting $I(k, l, p)$ to $I(k, cl, p)$ by a multiplier c. The *shadow*, as described in [10], of $I(k, l, p)$ at level cl is the product of component-wise preimage of the natural projection $\pi : \mathbb{Z}/lp\mathbb{Z} \to \mathbb{Z}/clp\mathbb{Z}$. Formally

$$\delta_{cl} I(k, l, p) = \bigcup_{v \in I(k,l,p)} \prod_{i=1}^{k} \pi^{-1}(v_i)$$

where $\pi : \mathbb{Z}/clp\mathbb{Z} \to \mathbb{Z}/lp\mathbb{Z}$ is the natural projection. Thus motivate the following lemma 4

Lemma 4. *Let $(l, p) = 1$ and $\pi : \mathbb{Z}/pl\mathbb{Z} \to \mathbb{Z}/p\mathbb{Z}$ be the natural projection. For any improper set $\{v_1, \ldots, v_k\}$ in $I(k, l, p)$, there is an improper set $\{u_1, u_2, \ldots, u_k\}$ in $I(k, l, p)$ and a number $u \in (\mathbb{Z}/pl\mathbb{Z})^\times$ such that $\pi(u_1) = 1$, i.e. $u_1 \equiv 1 \pmod p$, and*

$$\{uu_1, uu_2, \ldots, uu_k\} = \{v_1, \ldots, v_k\} \text{ as sets in } \mathbb{Z}/lp\mathbb{Z}$$

Proof. Since $(p, l) = 1$, the restriction $\pi|_{(\mathbb{Z}/pl\mathbb{Z})^\times} : (\mathbb{Z}/pl\mathbb{Z})^\times \to (\mathbb{Z}/p\mathbb{Z})^\times$ is surjective by the Chinese Remainder Theorem. Since $\{v_1, \ldots, v_k\}$ is an improper set, $\pi(v_1) \neq 0$, so $\pi(v_1) \in (\mathbb{Z}/p\mathbb{Z})^\times$. From surjectivity, there exists $u \in (\mathbb{Z}/pl\mathbb{Z})^\times$ with $\pi(u) = \pi(v_1)$, and pick $u_i := u^{-1}v_i$ for each i.

This makes $uu_i = v_i$ for all i, so $\{uu_1, \ldots, uu_k\} = \{v_1, \ldots, v_k\}$. Moreover,

$$\pi(u_1) = \pi(u)^{-1}\pi(v_1) = \pi(v_1)^{-1}\pi(v_1) = 1.$$

It remains to show that $\{u_1, \ldots, u_k\}$ is an improper set. For the gcd condition, since $u \in (\mathbb{Z}/pl\mathbb{Z})^\times$ we have $\gcd(u^{-1}, lp) = 1$, so multiplication by u^{-1} preserves all divisibility relations with respect to lp. Hence for each i,

$$\gcd(u_1, \ldots, \widehat{u}_i, \ldots, u_k, lp) = \gcd(v_1, \ldots, \widehat{v}_i, \ldots, v_k, lp) = 1,$$

where the last equality holds since $\{v_1, \ldots, v_k\}$ is an improper set.

For the second condition, pick any $t \in \mathbb{Z}/pl\mathbb{Z}$ and set $t' = tu^{-1}$ resembling lemma 3. Since $\{v_1, \ldots, v_k\}$ is improper, there exists some index i such that

$$\frac{1}{k+1} > \left\| \frac{t'v_i}{pl} \right\| = \left\| \frac{tu^{-1}v_i}{pl} \right\| = \left\| \frac{tu_i}{pl} \right\|$$

which is precisely the condition for $\{u_1, \ldots, u_k\}$. □

The lemma shows that every improper set in $I(k,l,p)$ is a unit multiple of an improper set whose first element projects to $1 \in \mathbb{Z}/p\mathbb{Z}$. In particular, it suffices to enumerate improper sets only among those with $\pi(u_1) = 1$, and it follows that

$$I(k,l,p) = \emptyset \iff \{\{u_1, \ldots, u_k\} \in I(k,l,p) : \pi(u_1) = 1\} = \emptyset.$$

This reduces the amount of computation needed by roughly a factor of p as it is enough to keep track of $I(k,l,p)/(\mathbb{Z}/p\mathbb{Z})^\times$.

In addition to this normalization, I improved the computational approach of Trakultongchai by refactoring the implementation to support parallel execution and by introducing more memory-efficient data structures. The new implementation remains faithful to the original algorithmic ideas but is significantly faster in practice.

Introducing intermediate steps

In this section, I write $I'(k,l,p) := I(k,l,p)/(\mathbb{Z}/p\mathbb{Z})^\times$. The cost of lifting $I'(k,l,p)$ to $I'(k,lc,p)$ is proportional to $|I'(k,l,p)| \cdot c^k$, since each improper set modulo lp has c^k candidate preimages modulo lcp to check. In particular, a lift with $c = 2$ costs $|I'(k,l,p)| \cdot 2^k$, whereas a lift with $c = 3$ costs $|I'(k,l,p)| \cdot 3^k$, making the $c = 2$ step significantly cheaper for large k.

Moreover, I observed that lifting with $c = 2$ can reduce the size of the improper set, i.e. $|I'(k,2l,p)| < |I'(k,l,p)|$ in practice. Thus, performing an additional $c = 2$ lift before a $c = 3$ lift can reduce the size of $I(k,l,p)$ enough to make the overall computation shorter, even accounting for the extra step. That is, the two-step strategy

$$I'(k,l,p) \xrightarrow{c=2} I'(k,2l,p) \xrightarrow{c=3} I'(k,6l,p)$$

can be cheaper than the direct lift

$$I'(k,l,p) \xrightarrow{c=6} I'(k,6l,p).$$

To put numbers to the observation, I have devised two runs to prove the case of $k = 8$. The first one follows Trakultongchai's method described in point 4.1 of [10]. The second one involve trying to lift with $c = 2$ twice and keep the smallest set. Specifically, the second method can be described as

(1) Generate $S_1 := I'(8,1,p)$
(2) Compute $S_2 := I'(8,2,p)$ by lifting S_1.
(3) If $|S_2| < |S_1|$, compute $S_4 := I'(8,4,p)$ by lifting S_2.
(4) Take T_1 as the smallest set among S_1, S_2 and S_4.
(5) Compute T_3 by lifting T_1 with $c = 3$.
(6) Compute T_9 by lifting T_3 with $c = 3$.

By running the two methods on a 10-core Apple M4 processor (Macbook Air 2025), the first method takes 18 seconds while the second methods uses 12 seconds,

This speedup can be dramatic in certain cases, for example in the case of $k = 9$. One idea is to generate $S_1 := I'(9,1,p)$, then lift it to $S_2 := I'(9,2,p)$ with $c = 2$ and lift it to $S_{10} := I'(9,10,p)$ with $c = 5$. My proposal is

(1) Generate $S_1 := I'(9,1,p)$
(2) Compute $S_2 := I'(9,2,p)$ by lifting S_1 with $c = 2$.
(3) Compute $S_4 := I'(9,4,p)$ by lifting S_2 with $c = 2$.
(4) Take T_1 to be the smaller set among S_2 and S_4.
(5) Compute T_3 by lifting T_1 with $c = 3$. At this point, T_3 can be either $I'(9,6,p)$ or $I'(9,12,p)$.

(6) Take U_1 to be the smaller set among T_1 and T_3.
(7) Compute U_5 by lifting U_1 with $c = 5$. At this point, $U_5 = I'(9, l, p)$ for some l such that $10 \mid l$.

For a specific number, the first method without the intermediate step could not compute $I'(9, 10, 53)$ within 15 minutes while it takes under 5 seconds with the second method. This opens up the possibility of lifting an improper set to a larger prime number and could help with the verification of the eleven runners conjecture.

Result

Rosenfeld wrote [5] to verify the conjecture for eight runners and Trakultongchai developed [9] to verify the conjecture of nine and ten runners. His computation uses 15 minutes for $k = 8$ and approximately 23 hours for $k = 9$. Building on the framework of [9], I developed [7], with assistance from AI-based tools, to improve performance and scalability.

Using the improvements proposed, I managed to complete the proof of $k = 8$ in 12 seconds and $k = 9$ in 181 seconds on a 10-core Apple M4 processor (Macbook Air 2025). The source of the computation can be found in [7]

For the twelve runners case, let

$$P_{11} := \left\{ \begin{array}{r} 23, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191, \\ 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251, 257, 263, \\ 269, 271, 277, 281, 283, 293, 307, 311, 313, 317, 331, 337, 347, \\ 349, 353, 359, 367, 373, 379, 383, 389, 397, 401, 409, 419, 421, \\ 431, 433, 439, 443, 449, 457, 461, 463, 467, 479, 487, 491, 499, \\ 503, 509, 521, 523, 541, 547, 557, 563, 569, 571, 577 \end{array} \right\}$$

Using the algorithm described in the previous section, I have verified that for every prime $p \in P_{11}$ there exists $l \leq 36$ such that the set $I(k, l, p)$ is empty. Equivalently, any counterexample to the 12-runner case of the Lonely Runners Conjecture must have speeds divisible by each prime in P_{11}. Consequently, if a counterexample exists, its product of speeds must be divisible by the product

$$\prod_{p \in P_{11}} p$$

Theorem 5. *The Twelve Lonely Runners Conjecture holds if the Eleven Lonely Runners Conjecture holds.*

Proof. If the conjecture holds for eleven runners, then the result of [6] asserts that it suffices to check integral solutions up to B_{11}.

As B_{11}, the bound needed to reduce the problem of twelve runners to eleven runners, is less than $e^{434.5}$ and $\prod_{p \in P_{11}} p > e^{435} > B_{11}$. There is no counterexample to the twelve runners case. □

Further work

The case of eleven runners remains unsolved. Since 11 is prime, the bottleneck of the computation is not at finding $I(k, 1, p)$ but at lifting $I(k, 1, p)$ to $I(k, 11, p)$. Similar arguments can potentially speed up calculation but will require more careful consideration.

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